

Galaxy color distribution estimation via dependent nonparametric mixtures

Stima della distribuzione del colore delle galassie via misture nonparametriche dipendenti

Antonio Canale, Riccardo Corradin and Bernardo Nipoti

Abstract We analyse a dataset drawn from the Sloan Digital Sky Survey first data release and consisting of the measurement of the $u - r$ color (i.e. difference between ultraviolet and red magnitude) of 24 312 galaxies. Galaxies are stratified by means of available covariate information on their luminosity and environment. An accurate estimate of the color distribution within each stratum provides interesting insight on how the population of galaxies changes across different luminosity and environment conditions. We propose and implement a Bayesian nonparametric mixture model for partially exchangeable data, obtained by normalising dependent superposed gamma processes. The structure of the proposed model allows for stratum-wise posterior inference while favouring borrowing of information across strata sharing same luminosity or same environment.

Abstract Analizziamo un dataset tratto dalla prima distribuzione di dati Sloan Digital Sky Survey e composto da misure di colore $u - r$ (cioè differenza tra magnitudine ultravioletta e rossa) di 24 312 galassie. Le galassie sono stratificate in base alle covariate luminosità e ambiente. Una stima accurata della distribuzione del colore all'interno di ogni strato permette di comprendere meglio come la popolazione di galassie cambi al variare di luminosità e ambiente. Proponiamo ed implementiamo un modello mistura bayesiano nonparametrico per dati parzialmente scambiabili, ottenuto normalizzando misture dipendenti di processi gamma. La struttura del modello proposto permette di fare inferenza specifica per ciascuno strato e, allo stesso tempo, favorisce scambio di informazione tra strati caratterizzati da stessa luminosità o stesso ambiente.

Antonio Canale

Dipartimento di Scienze Statistiche, Università degli studi di Padova, e-mail: `canale@stat.unipd.it`

Riccardo Corradin

DEMS, Università degli studi Milano-Bicocca, e-mail: `riccardo.corradin@unimib.it`

Bernardo Nipoti

School of Computer Science and Statistics, Trinity College Dublin, e-mail: `nipotib@tcd.ie`

Key words: Bayesian nonparametrics, dependent Dirichlet processes, mixture models, partially exchangeable data, galaxy surveys.

1 Astronomical application

Galaxies are known to be roughly composed by two populations, termed late-type and early-type galaxies. While this distinction originally refers to morphological properties of galaxies, the two populations can be approximately identified by their properties in terms of galaxy colors, such as, for instance, the $u - r$ color (i.e. difference between ultraviolet and red magnitude), with a bimodal distribution for the $u - r$ color reflecting the existence of two populations. Note that the definition of $u - r$ is such that larger values of $u - r$ correspond to redder colors. Considering that quantifying the $u - r$ color of galaxies is much simpler than assessing their morphological properties, considerable attention has been dedicated in the astronomical literature to the estimation of the color distribution and to the effect of available covariates on such distribution (see [1] and references therein). For instance, the $u - r$ color distribution is known to depend on both luminosity and galaxy environment. The luminosity is a measure of the light emitted by the galaxy in a given spectral band; the environment is quantified by measuring, for each galaxy, the density of neighbouring galaxies. In this paper we consider a dataset referring to $n = 24312$ galaxies, drawn from the Sloan Digital Sky Survey¹ (SDSS) first data release. Following [1], we consider a stratification of the SDSS dataset into 25 groups, based on five different levels of r -band luminosity and environment. Our goal is to accurately estimate the $u - r$ color distribution within each stratum so to provide interesting insight on the properties of the two main populations of galaxies and on how such properties change across strata.

Before describing our modelling framework, some notation is introduced. Let X_i , with $i = 1, \dots, n$, be the observed measurement of the $u - r$ color of a galaxy and let $Z_{1,i}$ and $Z_{2,i}$ be the related measures of its luminosity and environment. Both covariates are assumed discrete and take values in $\mathcal{L} = \{1, \dots, L\}$ and $\mathcal{G} = \{1, \dots, G\}$, respectively. Consistently with [1], here $L = G = 5$. This structure induces a stratification of the observations, which can be gathered into 25 distinct groups $\mathbf{X}^{(l,g)}$, with $l \in \mathcal{L}$ and $g \in \mathcal{G}$, where

$$\begin{aligned} \mathbf{X}^{(l,g)} &= \{X_i : Z_{1,i} = l, Z_{2,i} = g, 1 \leq i \leq n\} \\ &= \{X_1^{(l,g)}, \dots, X_{n_{l,g}}^{(l,g)}\}, \end{aligned}$$

and $\sum_{l \in \mathcal{L}} \sum_{g \in \mathcal{G}} n_{l,g} = n$.

¹ <https://www.sdss.org>, for project details and relevant references

2 Modelling framework

While homogeneity can be reasonably assumed within each group, distinct groups are characterized by heterogeneous luminosity and environment conditions. At the same time, any sound model for the SDSS data should account for the fact that, for any fixed l (or g), the groups of observations $\{\mathbf{X}^{(l,1)}, \dots, \mathbf{X}^{(l,G)}\}$ (or $\{\mathbf{X}^{(1,g)}, \dots, \mathbf{X}^{(L,g)}\}$) are characterized by common luminosity (or environment). This suggests that a model for partially exchangeable data could be adopted so to allow for stratum-wise posterior inference while favouring borrowing of information across strata sharing same luminosity or same environment.

2.1 Two-level GM-Dependent Dirichlet processes

The last two decades have witnessed a growing interest in models for partially exchangeable data. The introduction of the celebrated dependent Dirichlet process by MacEachern in [8, 9] paved the way to the definition of a plethora of nonparametric models inducing borrowing of information across groups of observations, see [3] for a review. Here we consider a class of mixture models with dependent random mixing probability measures, obtained by normalizing dependent superposed gamma completely random measures (CRM), in the spirit of [5]. Specifically, we consider the class of Griffiths-Milne dependent Dirichlet processes (GM-DDP), defined and studied in [6, 7] and we adapt it so to accommodate the specific nature of the SDSS data. Considering the two possible sources of common randomness across groups, namely common density or common environment, we refer to the resulting vector of random probability measures as to *two-level* GM-DDP. For an allied approach see [4], for a detailed introduction to CRMs see [2].

In order to define a vector of two-level GM-DDP, we consider independent sets of independent gamma CRMs, namely $\{\mu_{l,0} : l \in \mathcal{L}\}$, $\{\mu_{0,g} : g \in \mathcal{G}\}$ and $\{\mu_{l,g} : l \in \mathcal{L}, g \in \mathcal{G}\}$. We call $\nu_{l,0}$, $\nu_{0,g}$ and $\nu_{l,g}$ the corresponding Lévy intensities and assume that they are defined as

$$\begin{aligned} \nu_{l,0}(ds, d\theta) &= c(1-z)r \frac{e^{-s}}{s} ds P_0(d\theta), & l \in \mathcal{L} \\ \nu_{0,g}(ds, d\theta) &= c(1-z)(1-r) \frac{e^{-s}}{s} ds P_0(d\theta), & g \in \mathcal{G} \\ \nu_{l,g}(ds, d\theta) &= cz \frac{e^{-s}}{s} ds P_0(d\theta), & l \in \mathcal{L}, g \in \mathcal{G}, \end{aligned}$$

where $z, r \in [0, 1]$ and P_0 is a diffuse probability measure on some space Θ . For any $l \in \mathcal{L}$ and $g \in \mathcal{G}$, we define $\tilde{\mu}_{l,g}$ as the superposition of three independent CRMs, that is

$$\tilde{\mu}_{l,g} = \mu_{l,g} + \mu_{l,0} + \mu_{0,g}. \quad (1)$$

Marginally, the $\tilde{\mu}_{l,g}$'s are identically distributed gamma CRMs with Lévy intensity $\nu(ds, dx) = ce^{-s}/s ds P_0(dx)$. While $\mu_{l,g}$ only appears in the definition of $\tilde{\mu}_{l,g}$, the components $\mu_{l,0}$ and $\mu_{0,g}$ are shared by all the $\tilde{\mu}_{l,g}$ with common l and common g , respectively, and thus induce dependence across the components of the vector $(\tilde{\mu}_{1,1}, \tilde{\mu}_{1,2}, \dots, \tilde{\mu}_{L,G-1}, \tilde{\mu}_{L,G})$. In turn, by normalizing the CRMs $\tilde{\mu}_{l,g}$, we obtain a vector $\tilde{\mathbf{p}} = (\tilde{p}_{1,1}, \tilde{p}_{1,2}, \dots, \tilde{p}_{L,G-1}, \tilde{p}_{L,G})$ of two-level GM-DDP, whose elements are identically distributed Dirichlet processes $\tilde{p}_{l,g}$, with total mass $c > 0$ and base measure P_0 . We will use the notation $\tilde{\mathbf{p}} \sim \text{GM-DDP}(c, z, r, P_0)$. Each $\tilde{p}_{l,g}$ admits the convenient decomposition

$$\tilde{p}_{l,g} = w_{l,g} p_{l,g} + (1 - w_{l,g}) v_{l,g} p_{l,0} + (1 - w_{l,g})(1 - v_{l,g}) p_{0,g},$$

where $p_{l,g}$, $p_{l,0}$ and $p_{0,g}$ are independent Dirichlet processes distributed, respectively, as the processes obtained by normalizing $\mu_{l,g}$, $\mu_{l,0}$ and $\mu_{0,g}$. For every $l \in \mathcal{L}$ and $g \in \mathcal{G}$, the random weights $w_{l,g}$ and $v_{l,g}$ are given by $w_{l,g} = \mu_{l,g}(\boldsymbol{\theta}) / (\mu_{l,g}(\boldsymbol{\theta}) + \mu_{l,0}(\boldsymbol{\theta}) + \mu_{0,g}(\boldsymbol{\theta}))$ and $v_{l,g} = \mu_{l,0}(\boldsymbol{\theta}) / (\mu_{l,0}(\boldsymbol{\theta}) + \mu_{0,g}(\boldsymbol{\theta}))$. The vectors $\mathbf{w} = (w_{1,1}, w_{1,2}, \dots, w_{L,G-1}, w_{L,G})$ and $\mathbf{v} = (v_{1,1}, v_{1,2}, \dots, v_{L,G-1}, v_{L,G})$ are marginally independent and distributed according to the $(L \times G)$ -dimensional Beta distribution defined by [10]. More specifically,

$$\begin{aligned} \mathbf{w} &\sim \text{mult-Beta}_{L \times G}(cz, \dots, cz, c(1-z)) \\ \mathbf{v} &\sim \text{mult-Beta}_{L \times G}(c(1-z)r, \dots, c(1-z)r, c(1-z)(1-r)). \end{aligned}$$

Standard properties of gamma random variables conveniently imply independence between the random probability measures $p_{l,g}$, $p_{l,0}$, $p_{0,g}$ and the random weights $\mathbf{w}$ and $\mathbf{v}$.

2.2 Mixture model for partially exchangeable data

The components of $\tilde{\mathbf{p}}$ can be used as mixing random probability measures in order to define a vector of dependent mixture models. For every l, g and for any $i \in \{1, \dots, n_{l,g}\}$, let $\theta_i^{(l,g)} = (m_i^{(l,g)}, \tau_i^{(l,g)}) \in \mathbb{R} \times \mathbb{R}^+$ and $\boldsymbol{\theta}^{(l,g)} = (\theta_1^{(l,g)}, \dots, \theta_{n_{l,g}}^{(l,g)})$, and let $k(x; \boldsymbol{\theta})$ denote a univariate Gaussian kernel with parameters $\boldsymbol{\theta}$. The two-level GM-DDP mixture model for the SDSS data can be described by means of its hierarchical form as

$$\begin{aligned} (X_{i_{1,1}}^{(1,1)}, \dots, X_{i_{L,G}}^{(L,G)}) \mid \boldsymbol{\theta}^{(1,1)}, \dots, \boldsymbol{\theta}^{(L,G)} &\stackrel{\text{ind}}{\sim} \prod_{l=1}^L \prod_{g=1}^G k(x_{i_{l,g}}^{(l,g)}; \boldsymbol{\theta}_{i_{l,g}}^{(l,g)}) \\ \boldsymbol{\theta}_i^{(l,g)} \mid \tilde{\mathbf{p}} &\stackrel{\text{iid}}{\sim} \tilde{p}_{l,g} \quad l \in \mathcal{L}, g \in \mathcal{G}, i = 1, \dots, n_{l,g} \\ \tilde{\mathbf{p}} &\sim \text{GM-DDP}(c, z, r, P_0). \end{aligned}$$

It is interesting to interpret the role of the parameters z and r in modelling the SDSS data. The first one controls the weight given by the prior processes $\tilde{p}_{l,g}$ to the id-

iosyncratic component $p_{l,g}$ and thus affects the strength of the borrowing of information across groups. The latter parameter controls the relative weight given by $\tilde{p}_{l,g}$ to the components $p_{l,0}$ and $p_{0,g}$, thus determining which feature (common luminosity or common environment) is more impactful in inducing association across groups.

3 Results

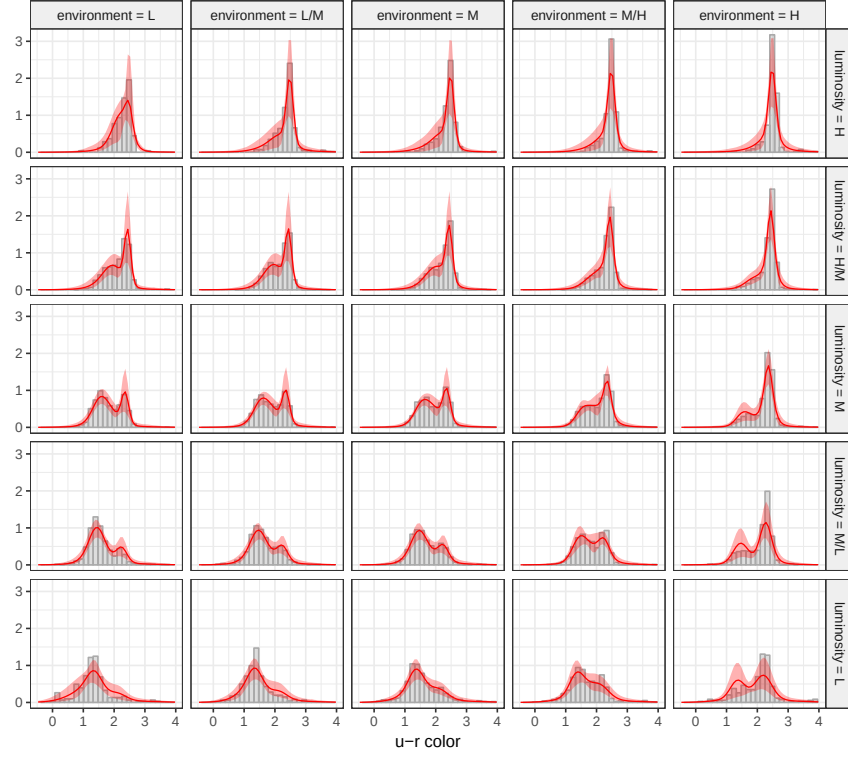


Fig. 1 Sloan Digital Sky Survey data, $u-r$ color distributions of galaxies stratified by means of co-variate information on luminosity (Z_1) and galaxy environment (Z_2). Each panel shows histogram, posterior estimated distribution (red line) and 90% posterior credible bands (red filled area). The groups correspond to *low*, *low/medium*, *medium*, *medium/high* and *high* values for luminosity and environment.

The model described in Section 2.2 is completed by setting $c = 1$ and assuming a normal-gamma base measure. More specifically, P_0 is defined so that if $(m, \tau) \sim P_0$, then $\tau \sim G(4, 1)$ and $m \mid \tau \sim N(2, 5\tau^{-1})$. The parameters z and r are both set equal to 0.5 so to formalise the idea that, a priori, idiosyncratic and common components

have the same weight and that, in turn, common luminosity and common environment are equally impactful in inducing borrowing of information across groups. The goal of our analysis consists in estimating the density function or the $u - r$ color of galaxies in each one of the 25 considered groups. To this end, the two-level GM-DDP mixture model was implemented and samples from the posterior distribution of $\tilde{\mathbf{p}}$ were obtained via a conditional Gibbs sampling algorithm, ran for 15 000 iterations, the first 5 000 of which were discarded as burn-in period. Convergence of the chain was assessed by visual investigation of the traceplots which provided no indication against it. Figure 1 displays the estimated densities, endowed with 90% posterior credible bands. A visual investigation of the posterior suggests that the borrowing of strength across strata has the effect of stressing the bimodality of the distribution of certain groups for which the histogram does not provide the same evidence. This effect can be particularly appreciated for groups with both low luminosity and high environment, such as, for example, the bottom right panel of the figure.

Acknowledgments

The authors are grateful to Michael Balogh for providing the dataset, and to Carlo Nipoti for fruitful comments on the astronomical problem.

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